Summary in Graph

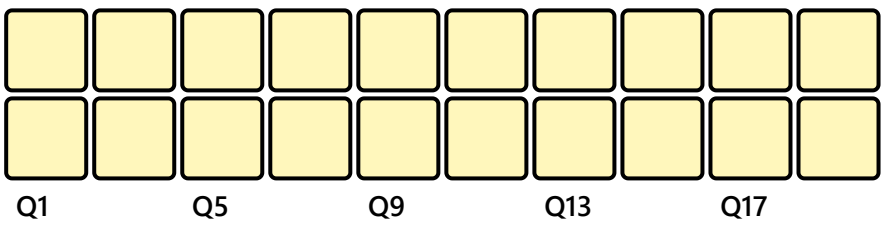
# Exam Summary (GO Classes GATE CS | C-Programming | Weekly Quiz 3)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 20 (15 + 5) |
| Total Marks: | 25 (15 + 10) |
| Exam Duration: | 60 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (45) out of 445 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Programming in C

What will be the output of the given C code?

```c
#include<stdio.h>
#include<stdlib.h>
int main(){
    int x, t[4] = {1,3,4,5};
    int *p, **pp;
    p = t;
    pp = &p;
    printf("%d ", *p);
    printf("%d ", **pp+1);
    printf("%d", *(*pp+1));
}
```

A. 1 3 4
B. 1 3 3
C. 1 2 3
D. 1 2 2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Which of the following is/are valid assignment(s) of the pointer for the given declaration? An assignment is valid if both pointers are of the same type (pointers are compatible) and get successfully compiled with no warning or error.

```c
int a[5]= {1,2};
int (*p)[5];
```

A. `p = a;`

B. `p = &a;`

C. `p = *a;`

D. `p = **a;`

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following declaration of pointer variable $p$.

```c
int (*p)[5];
```

If the initial value of $p$ is $1000$, then what will be the value of $p + 1$?

It is given that the system has $8$ bytes of address size and $4$ bytes of integer size.

A. $1001$
B. $1004$
C. $1020$
D. $1008$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Which of the following is NOT a valid C program? A valid program is a program having no compile-time errors.

```
Program 1
auto int a;
main ()
{
  auto int b;
  for (b = 0; b < 10; b++)
  {
    auto int a = b;
  }
}
```

```
Program 2
static int a;
main ()
{
  static int b;
  for (b = 0; b < 10; b++)
  {
    static int a = 1;
  }
}
```

```
Program 3
int a;
main ()
{
  static int b;
  for (b = 0; b < 10; b++)
  {
    static int a = 1;
  }
}
```

```
Program 4
extern int a;
main ()
{
  static int b;
  for (b = 0; b < 10; b++)
  {
    static int a = 1;
  }
}
```

A. Program 1
B. Program 2
C. Program 3
D. Program 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Which of the following is/are TRUE about storage classes in C programming

A. static local variables are NOT available to the linker.
B. static global variables are available to the linker.
C. extern local variables are available to the linker.
D. extern global variables are NOT available to the linker.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

The default storage class for functions in C language is –

A. Static
B. Auto
C. Extern
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

The default storage class for global variables in the C language is –

A. Static
B. Auto
C. Extern
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following declaration of struct.

```
    struct myst{
        char a[20];
        char *b;
        struct myst *p;
5.  }x[2] = {"GATE", "Overflow", x+1, "GO", "Classes", x}, *p = x;
```

What will be the output of the following print statement?

```
    printf("%s", p++->p->b++);
```

A. Overflow
B. Classes
C. verflow
D. lasses

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming and DS

Which of the following function declaration can be passed to the following array?

```
    char myArray[3][4];
```

A.
```
    void fun(char a[][], int size);
```

B.
```c
void fun(char a[][4], int size);
```

C.
```c
void fun(char [3][], int size);
```

D.
```c
void fun(char [][]a, int size);
```

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #10** Numerical Type | Award: 1 | Penalty: 0 | Programming in C

```c
char *strcollection[3] = {"GOClasses", "GATEOverflow", "IISc"};
```

For the above declaration, consider the base addresses of `strcollection`, `"GOClasses"`, `"GATEOverflow"` and `"IISc"` to be **1000**, **2000**, **3000** and **4000** respectively.

What will be the output of the below line?

```c
printf("%p", strcollection + 1);
```

Assume character is of one byte, integer is of two bytes, and address is of four bytes.

Your Answer: | Correct Answer: 1004 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

```c
char my_text[]= "2023 GATE";
void fun(int position)
{
    if (my_text[position] !=' ')
    {
        fun(position+1);
        printf("%c", my_text[position]);
    }
}
int main() {
    fun(0);
}
```

What will be the output of a given program ?

A. 2023
B. 3202
C. ETAG3202
D. ETAG

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #12** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

```
main()
{
    char s[]= "jjtd\0abc", *p;

    for (p = s; *p; p++)
    {
      --(*p);
    }

    printf("%s", s);
}
```

What will be the output of a given program?

A. iitb abc
B. iisc abc
C. iisc
D. iitb

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #13 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Which of the following(s) are true for following function

```
int mystery(int a) {
if(a == 256) return 3;
    return 1 + 2 * mystery(a*4); //line 3
}
```

A. $\text{mystery}(255)$ is an example of infinite recursion
B. if we replace line $3$ by following line return $1 +$ $\text{mystery}(\text{a*4})+$ $\text{mystery}(\text{a*4})$; then also output is same
C. Only possible outputs are $3, 7, 15, 31$
D. $\text{mystery}(\text{i})$ gives output if and only if $i$ is $1$ or multiple of $4$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #14 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be output of following program?

```
void xyz(int i)
{
    if(--i)
    {
        xyz(i++);
        printf("%d",i);
    }
}
main()
{
    xyz(5);
}
```

A. It is an infinite recursion since no base case
B. $2345$
C. $54321$
D. $1234$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #15

Numerical Type | Award: 1 | Penalty: 0 | Programming in C

What will be the output printed by $\text{mystery1}(0, 6)$?

```
void mystery1(int a, int b) {
    if (a <= b) {
        int m = (a + b) / 2;
        printf("%d", m);
        mystery1(a, m-1);
        mystery1(m+1, b);
    }
}
```

Your Answer: | Correct Answer: 3102546 | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #16

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the following program?

```
struct s{
    int i;
    struct s *p;
};
struct s arr[4] = { 7, arr+3,
    8, arr+2,
    9, arr,
    8, arr+1,
};
struct s *ap[] = {arr+3, arr+2, arr+1, arr};
struct s **pp =ap;
int main()
{
    printf("%d ", ap[++pp[2]->i-6]->i++);
    printf("%d ", pp[1]++->p++->i);
    printf("%d", ++arr[2].p->i);
}
```

A. 9 10 9
B. 7 8 10
C. 7 8 9
D. None of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #17

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Based on the struct alignment discussed in class,

What are the sizes of foo1, foo2, foo3 and foo4 respectively?

Sizes of primitive data types are given below.

```
sizeof(int) = 4 bytes
sizeof(short) = 2 bytes
sizeof(char) = 1 byte
```

```c
struct foo1 {
    int d1;
    char c1;
    int d2;
};

struct foo2 {
    int d1;
    char c1;
    int d2;
    char c2;
    short s;
};

struct foo3 {
    int d1;
    int d2;
    char c1;
    char c2;
    short s;
};

struct foo4 {
    char c1;
    int d1;
    short s;
    int d2;
    char c2;
};
```

A. 9, 12, 12, 12
B. 9, 12, 16, 20
C. 12, 16, 12, 20
D. 12, 20, 20, 20

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Mark all options which are likely to create problems with memory, i.e., run time error or has a memory leak.

Assume that malloc is successful in all cases.

```c
Program 1:
int *p = malloc(sizeof(int));
*p = 0;
free(p);
*p = 0;
```

```c
Program2:
int *p = malloc(sizeof(int));
*p = 0;
free(p);
p = 0;
```

```c
Program 3:
int *p = malloc(3*sizeof(int));
*p = 0;
p++;
free(p);
```

```c
Program 4:
int *p = malloc(sizeof(char));
*p = 0;
```

A. Program 1

B. Program $2$
C. Program $3$
D. Program $4$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following declarations of variables in a system having $4$ bytes for integers.

```
int a[5][4];
int (*p1)[2];
int (*p2)[2];
int (*t1)[4];
int (*t2)[4];
```

Suppose array a has base address of $1000$ and initial values of $p1, p2, t1$ and $t2$ are $1000, 1048, 1016$ and $1064$ respectively. What will be the value of $p2 - p1$ and $t2 - t1$ respectively?

A. $6\ 3$
B. $48\ 24$
C. $12\ 6$
D. $24\ 12$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Programming in C

```
int mystry (char* str) {
    if (*str == 0) {
        return 0;
    }
    return mystry (str+1) +1;
}
```

What will be the output of mystry("hello")?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

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